Son of Surya

GUIDE
R12
STUDENT & GLOBAL

STUDENTS | BEGINNER

Learning Solar: Career Roadmap

Solar job families, a six-level skills ladder, Suryamitra and other training, tools to learn, and how to build a portfolio and first role safely.

ITI	Suryamitra	Tools	Portfolio
START	TRAIN	DESIGN	PROOF

“Understand your solar before you buy your solar.”

English Edition · Version 1.0 · October 2026

Facts last verified: **7 October 2026** · sonofsurya.com

R12 About This Guide

Guide	**R12 — Learning Solar: A Student & Career Roadmap**
Category · Level	Student & Global · Beginner
For	School and college students, ITI and diploma learners, engineers and career-changers
Prerequisites	None
Related	The whole Son of Surya library — this guide shows the order to read it

The problem

Solar is one of India's fastest-growing industries, but students don't know which roles exist, which skills each needs, or how to start without expensive courses.

The decision

After reading, you can choose a target role, follow a step-by-step learning path, and build a portfolio that shows real skill.

01 What You Will Learn

- [ ] The main solar job families and what each person actually does
- [ ] A skills ladder from basics to professional
- [ ] Government-supported training such as Suryamitra
- [ ] Software and tools worth learning
- [ ] How to build a portfolio and find your first role safely

02 Why This Matters

Every rooftop, pump and plant needs people to sell, design, install, approve, finance and maintain it. Skills that are practical, safe and well documented are in demand across India and abroad.

03 Fundamentals — Job Families

Role	Does	Typical entry
Solar PV technician / installer	Installs structures, modules, wiring; tests and maintains	ITI/diploma + Suryamitra-type training
O&M technician	Cleans, inspects, troubleshoots, records	ITI/diploma; on-the-job

Solar pump technician	Installs and services farm pumps	ITI + Varunmitra-type training
Design engineer	Layouts, sizing, SLDs, yield estimates, structure coordination	Diploma/degree (electrical, mechanical, civil, energy)
Project engineer / manager	Runs sites, schedules, quality, safety, approvals	Degree + site experience
Sales / energy consultant	Surveys, proposals, customer education	Any background + strong technical basics
Liaison / approvals	DISCOM and portal paperwork	Detail-oriented; process knowledge
Finance / analyst	ROI models, loans, bids	Commerce/finance + solar basics

04 The Skills Ladder

Level	Learn	Son of Surya guides
1 Basics	Electricity units, bills, sun and shade	R1, R2, R3, R11, SOS-01
2 Products	Panels, inverters, batteries	R4, SOS-02, SOS-03, SOS-04
3 Design	Sizing, layout, net metering, approvals	SOS-06, SOS-08, SOS-09
4 Practice	Installation, wiring, safety, structure	SOS-23, SOS-24, R7, R8
5 Operation	Monitoring, maintenance, faults	SOS-20, SOS-21, SOS-22
6 Business	Quotes, finance, ROI, EPC, tenders	R5, SOS-16, SOS-17, SOS-26, SOS-27

05 How to Learn — Step by Step

1. Master the basics with free guides and practice the calculations.
2. Join a recognised training programme for hands-on safety and installation skills.
3. Learn one design tool and one spreadsheet model well.
4. Get site exposure (internship, assisting an installer) — always under supervision.
5. Document every project you help with (with permission) in a portfolio.
6. Specialise: rooftop, farm pumps, design, O&M, or business.

06 Indian Context — Training Options

VERIFY · MNRE SKILL PROGRAMMES

Suryamitra — solar PV technician training coordinated by the National Institute of Solar Energy (NISE), Gurugram, through training centres: about **3 months (600 hours)**, for candidates with at least 10th pass plus ITI in electrician/wireman/electronics/fitter/sheet-metal trades; assessed through the Skill Council for Green Jobs or other approved bodies. MNRE reported over 51,000 Suryamitras trained by December 2022. **Varunmitra** trains solar water pumping technicians. Source: MNRE skill development page; checked 7 Oct 2026 — check current batches and eligibility.

Other routes	Notes
ITI / polytechnic	Electrician and related trades are a strong base
Degrees	Electrical, mechanical, civil, energy, environmental engineering
Short courses	Check recognition, practical content and trainer experience before paying
Abroad	Many countries license electrical work separately — check local rules (R13)

07 Practical Application — Tools Worth Learning

Tool type	Examples (not endorsements)
Yield simulation	PVsyst, NREL SAM / PVWatts, Global Solar Atlas
Layout / CAD	AutoCAD or free CAD tools, SketchUp
Rooftop design platforms	Various web-based solar design tools
Spreadsheets	Sizing, BOQ, ROI models (SOS-17)
Field instruments	Multimeter, clamp meter, insulation tester, earth tester (under supervision)

08 Calculations — Practice Set

P1 From the basics

PROBLEM

A home uses 3,000 units a year in Tamil Nadu (3.4). First size? *Answer: 3,000 ÷ 365 ÷ 3.4 =* ***2.42 kWp***

P2 Design	
PROBLEM	String of 9 modules, Voc 50 V, β −0.27%/°C, coldest 8 °C. Max string Voc? *Answer: 50 × (1 + 0.0027 × 17) × 9 =* ***470.7 V***

P3 Business	
PROBLEM	Net cost ₹1,40,000; yearly saving ₹28,000. Simple payback? *Answer:* ***5 years***

09 Career Story

ILLUSTRATIVE EXAMPLE — not a real person

A student from Satara finished ITI (electrician), completed Suryamitra-type training, then assisted an installer for a year, photographing (with permission) each step and keeping test records. With a portfolio of 30 documented rooftop jobs and basic design skills, he became an O&M supervisor, and later began learning design tools to move into engineering.

10 Costs & Earnings

Training fees, tools and travel are investments — choose recognised programmes and avoid courses that promise guaranteed jobs or international placements for large fees. Pay varies widely by role, city and experience; ask working professionals.

11 Common Mistakes

Mistake	Better approach
Only theory, no site practice	Supervised hands-on work
Skipping safety training	Safety first — always
Paying for unrecognised “certificates”	Check recognition
No portfolio	Document your work

12 Protecting Yourself

- [ ] Never work on live electrical systems without training, supervision and PPE
- [ ] Verify training providers and placement promises
- [ ] Get offer letters and pay terms in writing

13 Your Roadmap

Step	Target date	Done
Read R1, R2, R3, SOS-01		
Solve 20 practice problems		
Join recognised training		
Learn one design tool		
Site experience (supervised)		
Portfolio of 10 documented jobs		

Son of Surya

LEARN WITH SON OF SURYA

Start with the free foundation guides and follow **@son.of.surya._** on Instagram for solar explainers.

14 FAQ

Q1 Do I need an engineering degree?
Not for technician roles; yes for most design engineering roles.

Q2 Can women build solar careers?
Yes — every role is open; choose supportive employers and training centres.

Q3 Is solar work only outdoors?
No — design, sales, finance and approvals are office roles.

Q4 Can I start my own business?
After experience, with licences, insurance and process (SOS-26).

15 Glossary

ITI — Industrial Training Institute.

NISE — National Institute of Solar Energy.

SCGJ — Skill Council for Green Jobs.

Suryamitra — solar PV technician programme.

Varunmitra — solar pump technician programme.

16 What To Do Next

R12 ✓ → R1 Basics → SOS-06 Sizing → SOS-23 Installation

17 Sources & Disclaimer

Sources: MNRE — Skill Development Programme page (Suryamitra, Varunmitra, NISE, SCGJ). Checked 7 Oct 2026.
Disclaimer: Educational guidance by Son of Surya. Programme details change — confirm with MNRE/NISE and training centres. Tools listed are examples, not endorsements. The career story is ILLUSTRATIVE.

Son of Surya

India’s structured solar knowledge library

Free foundation guides, practical calculators and honest, step-by-step knowledge — so every Indian home, farm and business can understand solar before buying it.

UNDERSTAND → CALCULATE → VERIFY → COMPARE → DECIDE → ACT

FREE TOOL

Son of Surya Solar Calculator

Enter your bills and location. See system size, yearly units and savings — with every assumption shown.

FREE TOOL

Solar Project Passport

Keep your solar project’s key numbers, documents and warranties in one place. An engineer-verified version is available.

FREE FOUNDATION GUIDES

- **R1** Electricity Basics for Solar
- **R2** How to Read Your Electricity Bill
- **R3** Sun, Site & Shade
- **SOS-01** Solar Starter Guide
- **R11** Solar Dictionary (English–Hindi)

SPECIALISED GUIDES — RELEASING IN BATCHES

- **Home** — rooftop planning, sizing, net metering
- **Kisan** — farm solar, pumps, schemes
- **Business & institutions** — shops, schools, societies
- **Professional** — wiring, installation, EPC

Powering the land. Protecting the water. Growing the future.

Made in India. Powered by the sun.

Licensed for the purchaser’s personal use. Please do not copy or redistribute. Educational content only — see the disclaimer inside. © 2026 Son of Surya.

Web sonofsurya.com **Instagram** @son.of.surya._ **WhatsApp** +49 160 4923611

Email sonofsurya15@gmail.com